

- (a) $\frac{d^2y}{dx^2} + 2y = 0$
 (b) $\frac{d^2y}{dx^2} + 2\frac{dy}{dx} + 2y = 0$
 (c) $\frac{d^2y}{dx^2} - 2\frac{dy}{dx} + 2y = 0$
 (d) $\frac{d^2y}{dx^2} + y = 0$

73. If $f(x) = ax - b$ and $g(x) = cx + d$ are such that $f[g(x)] = g[f(x)]$, then which one of the following holds?

- (a) $f(d) = g(b)$ (b) $f(b) + g(d) = 0$
 (c) $f(a) + g(c) = 2a$ (d) $f(d) + g(b) = 2d$

74. What is $\int_{-1}^1 (3 \sin x - \sin 3x) \cos^2 x dx$ equal to?

- (a) $-\frac{1}{4}$ (b) 0 (c) $\frac{1}{2}$ (d) $\frac{1}{4}$

75. What are the order and degree respectively of the differential equation $\left\{2 - \left(\frac{dy}{dx}\right)^2\right\}^{0.6} = \frac{d^2y}{dx^2}$?

- (a) 2, 2 (b) 2, 3 (c) 5, 2 (d) 2, 5

76. If $\frac{dy}{dx} = 2e^x y^3$, $y(0) = \frac{1}{2}$ then what is $4y^2(2 - e^x)$ equal to?

- (a) 1 (b) 2 (c) 3 (d) 4

77. Let $p = \int_a^b f(x) dx$ and $q = \int_a^b |f(x)| dx$. If $f(x) = e^{-x}$, then which one of the following is correct?

- (a) $p = 2q$ (b) $p = -q$
 (c) $4p = q$ (d) $p = q$

78. What is $\int_0^{\frac{\pi}{2}} \frac{a + \sin x}{2a + \sin x + \cos x} dx$ equal to?

- (a) $\frac{\pi}{4}$ (b) $\frac{\pi}{2}$ (c) 1 (d) 0

79. The non-negative values of b for which the function $\frac{16x^3}{3} - 4bx^2 + x$ has neither maximum nor minimum in the range $x > 0$ is

- (a) $0 < b < 1$ (b) $1 < b < 2$
 (c) $b > 2$ (d) $0 \leq b < 1$

80. Which one of the following is correct in respect of $f(x) = \frac{1}{\sqrt{|x| - x}}$ and $g(x) = \frac{1}{\sqrt{x - |x|}}$?

- (a) $f(x)$ has some domain and $g(x)$ has no domain
 (b) $f(x)$ has no domain and $g(x)$ has some domain
 (c) $f(x)$ and $g(x)$ have the same domain
 (d) $f(x)$ and $g(x)$ do not have any domain

Consider the following data for the next **two (02)** items that follow:

$$\text{Given that } \int \frac{3 \cos x + 4 \sin x}{2 \cos x + 5 \sin x} dx = \frac{\alpha x}{29} + \frac{\beta}{29} \ln |2 \cos x + 5 \sin x| + c$$

81. What is the value of α ?

- (a) 7 (b) 13 (c) 17 (d) 26

82. What is the value of β ?

- (a) 7 (b) 13 (c) 17 (d) 26

Consider the following data for the next **two (02)** items that follow:

$$\text{Let } f(x) = \frac{x}{\ln x}; (x > 1)$$

83. Consider the following statements:

- $f(x)$ is increasing in the interval (e, ∞)
- $f(x)$ is decreasing in the interval $(1, e)$
- $9 \ln 7 > 7 \ln 9$

Which of the statements given above are correct?

- (a) 1 and 2 only (b) 2 and 3 only
 (c) 1 and 3 only (d) 1, 2 and 3

84. Consider the following statements:

- $f''(e) = \frac{1}{e}$
- $f(x)$ attains local minimum value at $x = e$
- A local minimum value of $f(x)$ is e

Which of the statements given above are correct?

- (a) 1 and 2 only (b) 2 and 3 only
 (c) 1 and 3 only (d) 1, 2 and 3

Consider the following data for the next **two (02)** items that follow:

$$\text{Let } f(x) \text{ and } g(x) \text{ be two functions such that } g(x) = x - \frac{1}{x} \text{ and } fog(x) = x^3 - \frac{1}{x^3}.$$

85. What is $g[f(x) - 3x]$ equal to?

- (a) $x^3 - \frac{1}{x^3}$ (b) $x^3 + \frac{1}{x^3}$
 (c) $x^2 - \frac{1}{x^2}$ (d) $x^2 + \frac{1}{x^2}$